SummarIQ: Equation-Aware Summarization of Scientific Papers

AI-Powered Summaries Using Transformer Models

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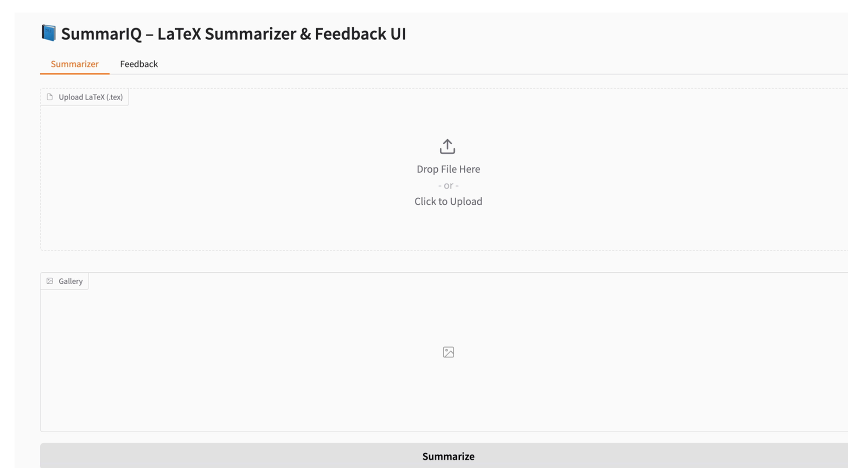
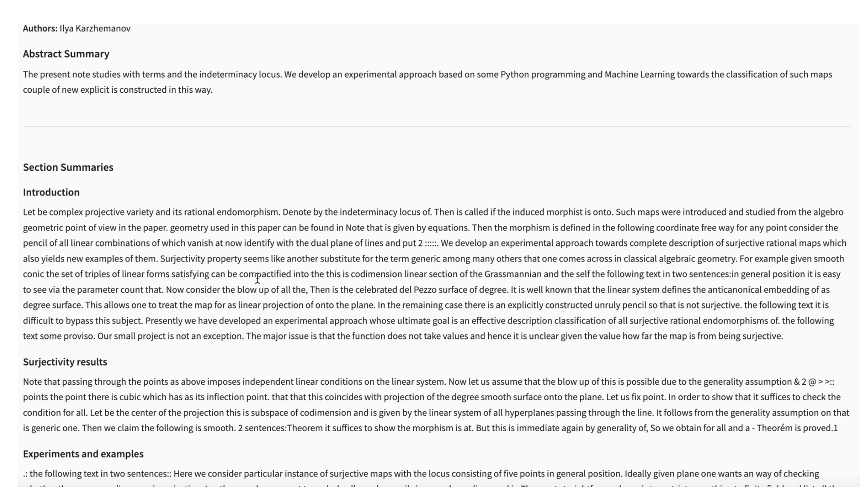
Why SummarIQ?

- Reading LaTeX research papers is slow and difficult due to complex structures, dense **mathematical expressions** and technical formatting.
- Existing summarisation tools focus mainly on plain text and fail to **preserve equations**, **section hierarchy**, and **scientific notation**.
- Researchers, students, and engineers need **quick**, **accurate summaries** that still retain the essential mathematical context.
- Modern transformer models like **T5** and **BART** are powerful, but they require careful preprocessing to work effectively on LaTeX documents.
- **SummarIQ** bridges this gap by providing an **equation-aware, transformer-based** summarization system tailored specifically for LaTeX scientific papers.

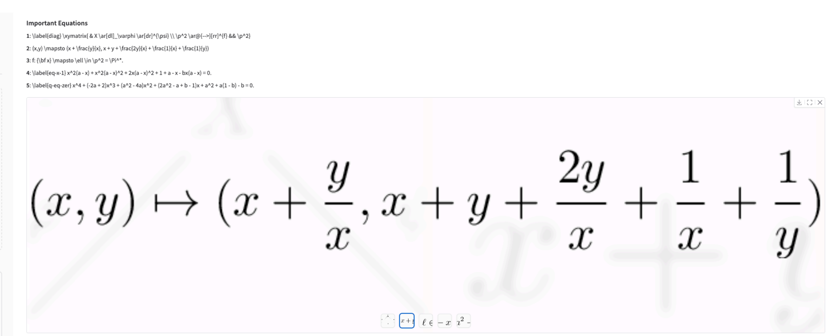
System Output Demonstration



- Fig. 1



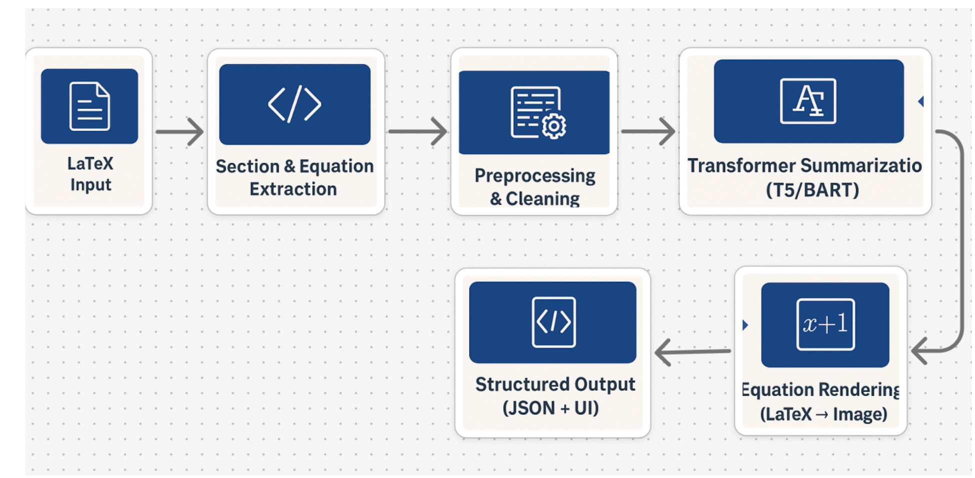
- Fig. 2



- Fig. 3

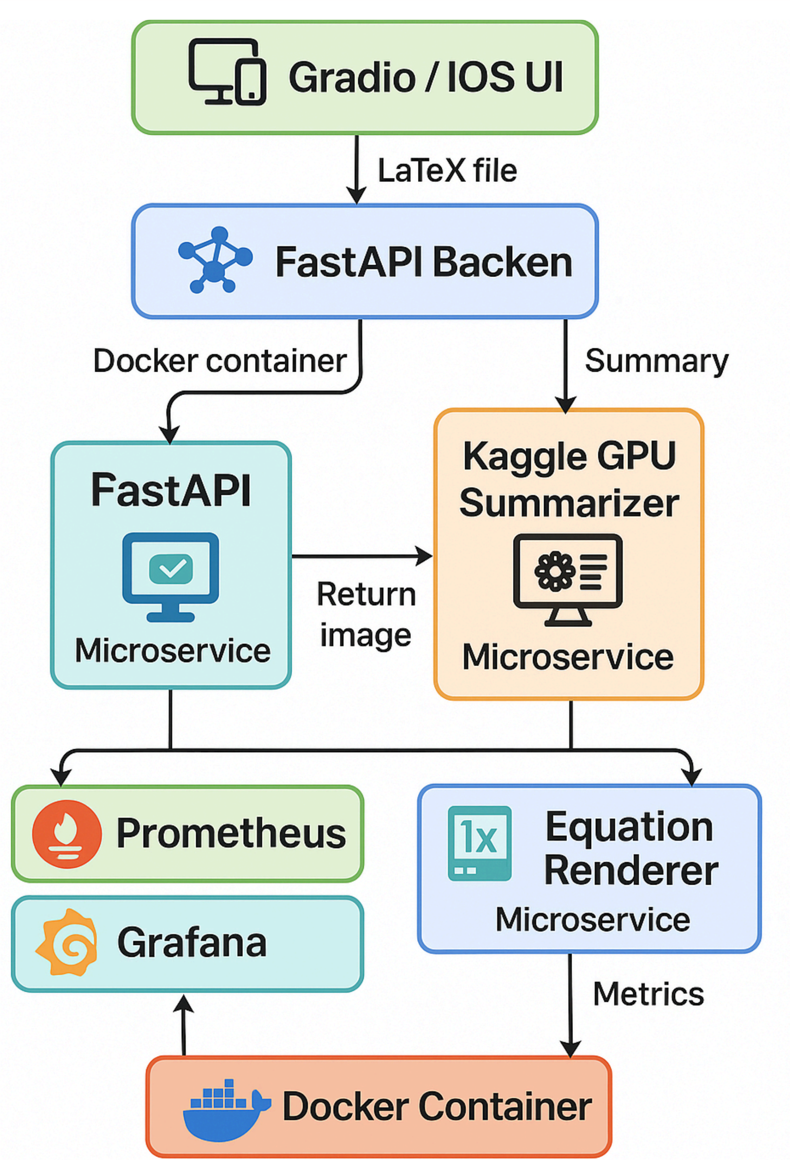
- **Fig. 1. Gradio User Interface**
  Shows the interactive UI where users upload LaTeX files and view generated summaries.
- **Fig. 2. Section-Level Summary Output**
  Displays the transformer-generated scientific summaries produced by the T5/BART model.
- **Fig. 3. Rendered Equation Example**
  Demonstrates how LaTeX equations are converted into clean PNG images for improved readability.

Model Pipeline



- **Structured Output Generation**
  Summaries and metadata are packaged into a structured JSON response.
- **LaTeX Extraction & Preprocessing**
  Sections, subsections, and mathematical equations are extracted and cleaned from the **.tex** document.
- **Transformer-Based Summarization**
  Clean scientific text is summarized using T5/BART to generate concise, section-level summaries.
- **LaTeX Extraction & Preprocessing**
  Important LaTeX equations are converted into readable images and displayed in the Gradio/iOS interface.

Architectural Diagram / Workflow



- **FastAPI Backend acts as the central controller**, receiving LaTeX input, extracting sections and equations , performing preprocessing, and coordinating communication between the summarizer and renderer microservices.
- **A remote Kaggle GPU Summarization Service** executes the heavy transformer models (T5/BART) , enabling fast, GPU-accelerated generation of scientific summaries without requiring local high-end hardware.
- **The Equation Renderer Microservice** processes mathematical expressions extracted from LaTeX and converts them into clean, high-resolution PNG/SVG images, making complex equations easier for users to read and interpret.
- **All system components run inside Docker containers**, providing isolation, consistency, reproducibility, and simplified multi-service deployment across different environments.
- **Prometheus and Grafana form the monitoring layer**, capturing API latency, request volume, and custom metrics like ***summariq_feedback_total*** to support real-time performance visualization and system health tracking.

Implementation Results

- **T5-small and BART-large models** successfully generated concise section-level summaries with consistent scientific coherence.
- **Equation Renderer** produced high-quality PNG equation images for all extracted math expressions, improving readability for non-LaTeX users.

Metric	Value
ROUGE-1	39.02
ROUGE-2	15.72
ROUGE-L	24.15
Latency	< 5 sec
Equation Success	90%

GitHub Repository

LinkedIn Github Kaggle